

Team 17 Requirements Report

CS 200 Project 1: Chocoholics Anonymous (ChocAn)

1. Header Table

Name	CWID	Email	Hours Contributed
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2. Task Distribution Table

Name	Tasks Performed	Percentage
Trey Pierce	Actors list, GitHub repository, provider use case descriptions	20%
Adam Davis	Accounting use case descriptions	20%
Clark Berry	Operator and Acme use case descriptions, submission	20%
Samuel Harvey	Use case diagram	20%
Richard Norman	Glossary, solution paragraph, report assembly	20%
Total		100%

3. Solution Paragraph

ChocAn (Chocoholics Anonymous) is an organization that helps people who are addicted to chocolate. Members pay a monthly fee and can see health care providers as often as they need. Our solution is a data processing system for the ChocAn Data Center. Providers use it to log in, validate members, bill for services, and request the Provider Directory. Operators use it to add, update, and delete member and provider records. At midnight every Friday, a timer starts the main accounting procedure, which reads the week's services and creates member reports, provider reports, a summary report, and an EFT record. The ChocAn Manager can also request a member, provider, or summary report at any time. Each evening at 9 PM, Acme Accounting Services updates member status. The actors are the Provider, the Operator, the Timer, the ChocAn Manager, and Acme Accounting Services. We will write the system in Java. As the contract requires, terminal input is simulated with the keyboard, terminal output appears on the screen, and every report and the Provider Directory are written to their own files instead of being sent by email.

We assume the Acme update arrives on schedule each evening, that only Acme can change member status, that new members start as active, and that the weekly run writes zero totals when no services were billed. We also assume that when a provider number is not accepted, the system asks for it again; that a suspended member cannot be billed; and that when the service name shown is wrong, the provider enters the service code again. This report includes the header table (Section 1), the task distribution table (Section 2), the glossary (Section 4), the use case diagram (Section 5), and a description for each use case (Section 6). Our team members are Trey Pierce, Adam Davis, Clark Berry, Samuel Harvey, and Richard Norman. Clark Berry is the submitter.

4. Glossary

ChocAn (Chocoholics Anonymous): An organization that helps people who are addicted to chocolate.

Member: A person who pays ChocAn a monthly fee and receives services for chocolate addiction.

Provider: A health care professional who gives services to ChocAn members.

Operator: The actor who adds, updates, and deletes member and provider records at the ChocAn Data Center.

Timer: The actor that starts the main accounting procedure at midnight every Friday.

ChocAn Manager: The actor who can request a member report, provider report, or summary report at any time.

Acme Accounting Services: An outside company that records membership fee payments, suspends members whose fees are overdue, and reinstates members who pay. Its computer updates ChocAn member records each evening at 9 PM.

Use Case: A task an actor performs with the system, such as Bill for Service.

Actor: A person or outside system that interacts with the ChocAn system.

Include: A relationship where one use case always runs another use case as part of its steps.

Extend: A relationship where a use case adds optional or exception steps to another use case.

Main Success Scenario: The numbered steps a use case follows when everything goes right.

Extension: An alternative or error path that replaces a step in the main success scenario.

ChocAn Data Center: The location of the ChocAn computer that stores member and provider records and runs the software.

Provider Terminal: The ChocAn terminal a provider uses to log in, validate members, and bill for services.

Member Card: A plastic card with the member's name and nine-digit member number, which are also stored on a magnetic strip.

Member Number: A unique nine-digit number that identifies each ChocAn member.

Provider Number: A unique nine-digit number that identifies each provider.

Member Status: Shows whether a member is active or suspended. A member is suspended when fees are owed.

Service Code: A six-digit number that identifies a ChocAn service and its fee.

Consultation: One service given by a provider to a member.

Fee: The amount ChocAn pays a provider for a service.

Service Record: The record written to disk when a provider bills for a service. It holds the current date and time, the date of service, the provider number, the member number, the service code, and any comments.

Provider Directory: An alphabetical list of every ChocAn service with its service code and fee.

Main Accounting Procedure: The weekly process that reads the week's services and creates the member reports, provider reports, summary report, and EFT record.

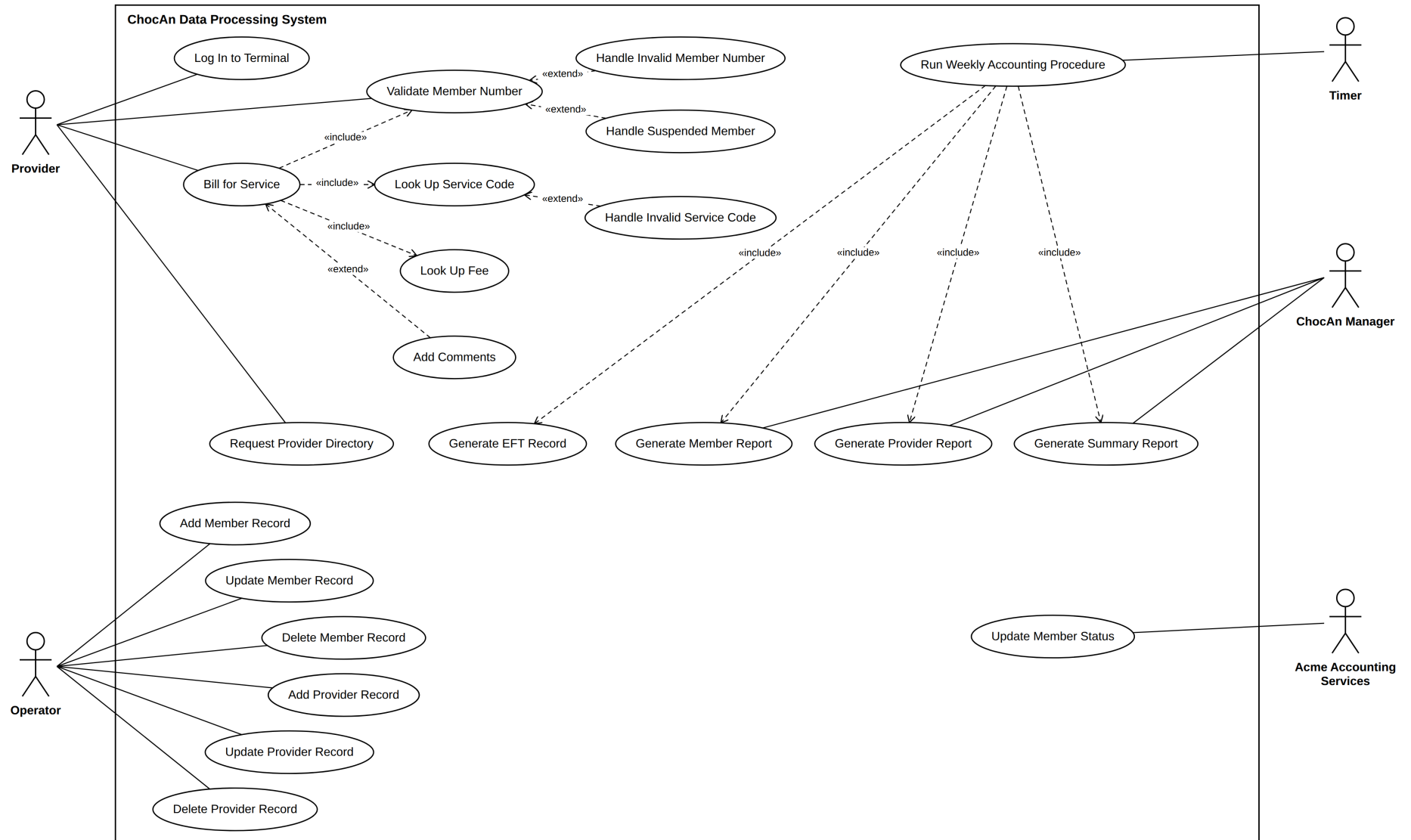
Member Report: A weekly report for each member who received services. It lists the member's name, number, and address, and for each service the date, the provider name, and the service name.

Provider Report: A weekly report for each provider who billed ChocAn. It lists the services the provider gave that week and the total fee owed to the provider.

Summary Report: A weekly report for the accounts payable manager. It lists every provider to be paid, their number of consultations, and their total fee, followed by the overall totals.

EFT (Electronic Funds Transfer) Record: A file with each provider's name, provider number, and amount to be paid. Banking computers use it to pay providers.

5. Use Case Diagram



6. Use Case Descriptions

Provider Use Cases

Use Case: Log In to Terminal

Context: The provider turns on the terminal and enters a nine-digit provider number.

Actors: Provider

Main Success Scenario:

1. Provider turns on the terminal.
 - 1.1. System asks for the provider number.
2. Provider enters the provider number.
3. System accepts the number and turns on the billing functions.
4. Use case ends.

Extensions:

- 3a. The provider number is not accepted. System asks for the number again and returns to step 2.

Use Case: Request Provider Directory

Context: The provider asks for an alphabetical list of service names, service codes, and fees.

Actors: Provider

Main Success Scenario:

1. Provider requests the Provider Directory.
2. System gets every service name, service code, and fee.
 - 2.1. System sorts the entries alphabetically by service name.
3. System writes the Provider Directory to a file.
4. Use case ends.

Extensions:

N/A

Use Case: Bill for Service

Context: The provider bills ChocAn after giving a health care service to a member.

Actors: Provider

Main Success Scenario:

1. Provider swipes the member card or enters the nine-digit member number.
2. System validates the member number (include Validate Member Number).
 - 2.1. Display shows "Validated".
3. Provider enters the date the service was provided (MM-DD-YYYY).
4. Provider looks up the six-digit service code in the Provider Directory and enters it.
5. System looks up the code and shows the service name, up to 20 characters (include Look Up Service Code).
6. Provider confirms that the service name is the service that was provided.
7. System writes the service record to disk.
 - 7.1. Current date and time (MM-DD-YYYY HH:MM:SS).
 - 7.2. Date of service (MM-DD-YYYY).
 - 7.3. Provider number (9 digits).
 - 7.4. Member number (9 digits).
 - 7.5. Service code (6 digits).
 - 7.6. Comments (optional, up to 100 characters).
8. System looks up the fee and shows it on the terminal (include Look Up Fee).
9. Use case ends.

Extensions:

- 2a. The member number is invalid. See Handle Invalid Member Number. Use case ends.
- 2b. The member is suspended. See Handle Suspended Member. Use case ends.
- 5a. The service code does not exist. See Handle Invalid Service Code. System returns to step 4.
- 6a. The service name shown is not the service that was provided. System returns to step 4.
- 6b. After confirming the service, the provider chooses to enter comments. See Add Comments. System continues at step 7.

Use Case: Validate Member Number

Context: The system checks a member number against ChocAn Data Center records and shows the result on the terminal's one-line display.

Actors: Provider

Main Success Scenario:

1. Provider swipes the member card or enters the nine-digit member number, or the number is passed from step 1 of Bill for Service.
2. Terminal dials the ChocAn Data Center.
3. ChocAn Data Center computer checks the member number.
4. Display shows "Validated".
5. If Bill for Service started this use case, control returns to step 3 of Bill for Service. Otherwise, the use case ends.

Extensions:

- 3a. The number is not a valid member number. See Handle Invalid Member Number. Use case ends.
- 3b. The member status is suspended. See Handle Suspended Member. Use case ends.

Use Case: Handle Invalid Member Number

Context: The system reports that the number entered is not a valid ChocAn member number.

Actors: Provider (through Validate Member Number)

Main Success Scenario:

1. System finds that the member number is not valid.
2. System shows "Invalid number" on the one-line display.
3. The member is not validated, and control returns to Validate Member Number, which ends (extension 3a).

Extensions:

N/A

Use Case: Handle Suspended Member

Context: The member number is valid, but the member status is suspended because membership fees have not been paid for at least a month.

Actors: Provider (through Validate Member Number)

Main Success Scenario:

1. System finds that the member status is suspended.
2. System shows "Member suspended" on the one-line display.
3. The member is not validated, no service can be billed, and control returns to Validate Member Number, which ends (extension 3b).

Extensions:

N/A

Use Case: Look Up Service Code

Context: The system finds the service name for the six-digit service code the provider entered, so the provider can confirm it.

Actors: Provider (through Bill for Service)

Main Success Scenario:

1. System receives the six-digit service code.
2. System looks up the service name for that code.
3. System shows the service name (up to 20 characters) and asks the provider to confirm it is the service that was provided.
4. Control returns to step 6 of Bill for Service.

Extensions:

- 2a. The service code does not exist. See Handle Invalid Service Code. Use case ends.

Use Case: Handle Invalid Service Code

Context: The system reports that the six-digit code the provider entered does not match any ChocAn service.

Actors: Provider (through Look Up Service Code)

Main Success Scenario:

1. System finds that the service code does not exist.
2. System shows an error message.
3. Control returns to Look Up Service Code, which ends. Bill for Service then returns to step 4 so the provider can enter the service code again.

Extensions:

N/A

Use Case: Look Up Fee

Context: The system finds the fee for the confirmed service and shows it on the provider's terminal.

Actors: Provider (through Bill for Service)

Main Success Scenario:

1. System receives the confirmed service code.
2. System looks up the fee for that service.
3. System shows the fee (up to \$999.99) on the provider's terminal.
4. Control returns to step 9 of Bill for Service.

Extensions:

N/A

Use Case: Add Comments

Context: The provider enters optional comments about the service before the service record is written to disk.

Actors: Provider (through Bill for Service)

Main Success Scenario:

1. Provider chooses to enter comments about the service provided.
2. Provider enters comments (up to 100 characters).
3. System puts the comments in the comments field of the service record.
4. Control returns to step 7 of Bill for Service.

Extensions:

N/A

Operator Use Cases

Use Case: Add Member Record

Context: The operator creates a new member record.

Actors: Operator

Main Success Scenario:

1. Operator selects Add Member Record.
2. System asks for the member data.
 - 2.1. Name (25 characters).
 - 2.2. Member number (9 digits).
 - 2.3. Street address (25 characters).
 - 2.4. City (14 characters).
 - 2.5. State (2 letters).
 - 2.6. ZIP code (5 digits).
3. Operator enters the data.
4. System checks the fields and confirms the member number is not in use.
5. System saves the record and sets the member status to active.
6. System confirms the record was created and returns to the main menu.

Extensions:

- 4a. The member number is in use. System shows an error, saves nothing, and returns to step 2.
- 4b. A field is invalid. System shows an error and returns to that field in step 3.

Use Case: Update Member Record

Context: The operator changes the details of an existing member.

Actors: Operator

Main Success Scenario:

1. Operator selects Update Member Record.
2. Operator enters the member number.
3. System shows the record.
4. Operator edits the name, street address, city, state, or ZIP code.
5. System checks the changes and saves the record.
6. System confirms the update and returns to the main menu.

Extensions:

- 3a. The member number is not found. System shows an error and returns to step 2.
- 4a. Operator tries to edit the member number. System refuses, because the member number is the record key, and returns to step 4.
- 4b. Operator tries to change the member status. System refuses, because Acme Accounting Services sets member status each evening at 9 PM, and returns to step 4.
- 5a. A field is invalid. System shows an error and returns to step 4.

Use Case: Delete Member Record

Context: The operator removes a member who has resigned.

Actors: Operator

Main Success Scenario:

1. Operator selects Delete Member Record.
2. Operator enters the member number.
3. System shows the record and asks for confirmation.
4. Operator confirms.
5. System deletes the record.
6. System confirms the deletion and returns to the main menu.

Extensions:

- 3a. The member number is not found. System shows an error and returns to step 2.
- 4a. Operator cancels. System changes nothing and returns to the main menu.

Use Case: Add Provider Record

Context: The operator creates a new provider record.

Actors: Operator

Main Success Scenario:

1. Operator selects Add Provider Record.
2. System asks for the provider data.
 - 2.1. Name (25 characters).
 - 2.2. Provider number (9 digits).
 - 2.3. Street address (25 characters).
 - 2.4. City (14 characters).
 - 2.5. State (2 letters).
 - 2.6. ZIP code (5 digits).
3. Operator enters the data.
4. System checks the fields and confirms the provider number is not in use.
5. System saves the record.
6. System confirms the record was created and returns to the main menu.

Extensions:

- 4a. The provider number is in use. System shows an error, saves nothing, and returns to step 2.
- 4b. A field is invalid. System shows an error and returns to that field in step 3.

Use Case: Update Provider Record

Context: The operator changes the details of an existing provider.

Actors: Operator

Main Success Scenario:

1. Operator selects Update Provider Record.
2. Operator enters the provider number.
3. System shows the record.
4. Operator edits the name, street address, city, state, or ZIP code.
5. System checks the changes and saves the record.
6. System confirms the update and returns to the main menu.

Extensions:

- 3a. The provider number is not found. System shows an error and returns to step 2.
- 4a. Operator tries to edit the provider number. System refuses, because the provider number is the record key and is used in the EFT record, and returns to step 4.
- 5a. A field is invalid. System shows an error and returns to step 4.

Use Case: Delete Provider Record

Context: The operator removes a provider who has left ChocAn.

Actors: Operator

Main Success Scenario:

1. Operator selects Delete Provider Record.
2. Operator enters the provider number.
3. System shows the record and asks for confirmation.
4. Operator confirms.
5. System deletes the record.
6. System confirms the deletion and returns to the main menu.

Extensions:

- 3a. The provider number is not found. System shows an error and returns to step 2.
- 4a. Operator cancels. System changes nothing and returns to the main menu.

Timer and ChocAn Manager Use Cases

Use Case: Run Weekly Accounting Procedure

Context: At midnight on Friday, the main accounting procedure processes the week's record of services provided, calculates total fees, and creates all required files and reports.

Actors: Timer

Main Success Scenario:

1. At midnight on Friday, the Timer starts the main accounting procedure.
2. System opens and reads the week's file of recorded services.
3. For each service, the system reads the stored fields:
 - 3.1. Current date and time (MM-DD-YYYY HH:MM:SS).
 - 3.2. Date of service (MM-DD-YYYY).
 - 3.3. Provider number (9 digits).
 - 3.4. Member number (9 digits).
 - 3.5. Service code (6 digits).
 - 3.6. Comments (optional, up to 100 characters).
4. System groups and sorts all service data by member, by provider, and by overall totals.
5. System runs the included use cases to create all required files:
 - 5.1. Generate Member Report (one file per member who received services).
 - 5.2. Generate Provider Report (one file per provider who billed ChocAn).
 - 5.3. Generate Summary Report for the accounts payable manager.
 - 5.4. Generate EFT Record with the amount to pay each provider.
6. System shows and logs a message that the weekly accounting is complete, and the procedure ends.

Extensions:

- 2a. The weekly service file is empty or missing.
 - 2a.1. System logs "No service data found for the specified period".
 - 2a.2. System skips the member and provider reports and continues at step 5.3, where the summary report is created with zero totals and the EFT record says "No transfers scheduled".
- 3a. A record in the service file is corrupted or badly formatted.
 - 3a.1. System logs the record in an error log file.
 - 3a.2. System skips the record and continues with the remaining records.
- 5a. The procedure crashes before all files are created.
 - 5a.1. System logs the failure.
 - 5a.2. System restarts the procedure at step 2.
- 6a. An included use case reported an error.
 - 6a.1. System shows and logs that the weekly accounting finished with errors, and the procedure ends.

Use Case: Generate Member Report

Context: Creates a report file for every member who received services during the week. The file is saved instead of being sent by email.

Actors: ChocAn Manager; Timer (through Run Weekly Accounting Procedure)

Main Success Scenario:

1. The report is started by Run Weekly Accounting Procedure or by the ChocAn Manager.
2. System reads the weekly service records and groups them by member number.
3. For each member who saw a provider during the week, the system creates a new file named with the member's name followed by the report date.
4. System writes the member's information to the file:
 - 4.1. Member name (25 characters).
 - 4.2. Member number (9 digits).
 - 4.3. Member street address (25 characters).
 - 4.4. Member city (14 characters).
 - 4.5. Member state (2 letters).
 - 4.6. Member ZIP code (5 digits).
5. System sorts the member's services by date of service (MM-DD-YYYY).
6. For each service, the system writes:
 - 6.1. Date of service (MM-DD-YYYY).
 - 6.2. Provider name (25 characters).
 - 6.3. Service name (20 characters).
7. System saves and closes the report file.
8. System continues at step 5.2 of Run Weekly Accounting Procedure, or returns to the main menu if the ChocAn Manager started the report.

Extensions:

- 4a. Member address data or a required field is missing.
 - 4a.1. System writes "N/A" for the missing field and logs a warning.
 - 4a.2. System finishes creating the file.

Use Case: Generate Provider Report

Context: Creates a report file for each provider who billed ChocAn during the week, listing each consultation and the total fee owed.

Actors: ChocAn Manager; Timer (through Run Weekly Accounting Procedure)

Main Success Scenario:

1. The report is started by Run Weekly Accounting Procedure or by the ChocAn Manager.
2. System reads the weekly service data and groups it by provider number.
3. For each provider who billed ChocAn during the week, the system creates a new file named with the provider's name followed by the report date.
4. System writes the provider's information:
 - 4.1. Provider name (25 characters).
 - 4.2. Provider number (9 digits).
 - 4.3. Provider street address (25 characters).
 - 4.4. Provider city (14 characters).
 - 4.5. Provider state (2 letters).
 - 4.6. Provider ZIP code (5 digits).
5. System lists each service in the order the computer received the data:
 - 5.1. Date of service (MM-DD-YYYY).
 - 5.2. Date and time data were received by the computer (MM-DD-YYYY HH:MM:SS).
 - 5.3. Member name (25 characters).
 - 5.4. Member number (9 digits).
 - 5.5. Service code (6 digits).
 - 5.6. Fee to be paid (up to \$999.99).
6. System writes the summary at the end of the report:
 - 6.1. Total number of consultations with members (3 digits).
 - 6.2. Total fee for the week (up to \$99,999.99).
7. System saves and closes the report file.
8. System continues at step 5.3 of Run Weekly Accounting Procedure, or returns to the main menu if the ChocAn Manager started the report.

Extensions:

- 6a. The provider's total fee is more than \$99,999.99.
 - 6a.1. System flags the total in the report and logs the account for ChocAn Manager review.

Use Case: Generate Summary Report

Context: Creates the accounts payable summary report, listing every provider to be paid, the consultation totals, and the total amount ChocAn will pay.

Actors: ChocAn Manager; Timer (through Run Weekly Accounting Procedure)

Main Success Scenario:

1. The report is started by Run Weekly Accounting Procedure or by the ChocAn Manager.
2. System processes all of the week's service records.
3. System creates a new summary report file for the accounts payable manager.
4. System lists every provider to be paid that week:
 - 4.1. Provider name.
 - 4.2. Number of consultations by that provider.
 - 4.3. Total fee owed to that provider for the week.
5. System adds the totals at the end of the report:
 - 5.1. Total number of providers who provided services.
 - 5.2. Total number of consultations.
 - 5.3. Overall fee total to be paid by ChocAn.
6. System saves the report file.
7. System continues at step 5.4 of Run Weekly Accounting Procedure, or returns to the main menu if the ChocAn Manager started the report.

Extensions:

- 4a. No services were given during the week.
 - 4a.1. System creates the report with all totals set to zero (0 providers, 0 consultations, \$0.00 total fee).

Use Case: Generate EFT Record

Context: Creates the Electronic Funds Transfer (EFT) file with provider payment details so banking computers can pay providers.

Actors: Timer (through Run Weekly Accounting Procedure)

Main Success Scenario:

1. System finds all providers who are owed payment for services billed this week.
2. System creates or replaces the EFT data file on disk.
3. For each provider to be paid, the system writes a line with:
 - 3.1. Provider name (25 characters).
 - 3.2. Provider number (9 digits).
 - 3.3. Amount to be transferred (total fee for the week).
4. System saves the file to disk for the banking computers.
5. System continues at step 6 of Run Weekly Accounting Procedure.

Extensions:

- 1a. No providers billed ChocAn this week.
 - 1a.1. System creates an EFT file that says "No transfers scheduled".
 - 1a.2. System continues at step 6 of Run Weekly Accounting Procedure.
- 3a. The file cannot be written because of a disk or permission error.
 - 3a.1. System logs "EFT file creation failed" for the ChocAn Manager.
 - 3a.2. System continues at step 6 of Run Weekly Accounting Procedure.

Acme Accounting Services Use Cases

Use Case: Update Member Status

Context: Each evening at 9 PM, the Acme computer updates ChocAn member status based on fee payments.

Actors: Acme Accounting Services

Main Success Scenario:

1. At 9 PM, the Acme computer connects to the ChocAn Data Center.
2. Acme sends the list of member status changes.
 - 2.1. Members to suspend because fees are overdue.
 - 2.2. Members to reinstate because they have paid.
3. For each change, the system finds the member record by member number.
4. System sets the member status to suspended or active.
5. System saves the records and confirms the update to Acme.
6. The update ends.

Extensions:

- 3a. The member number is not found. System logs the error and moves to the next change.